Bambang Suhartono

2026-Paper 3 - DSR ITSP - Bambang S - Template IJIES - INASS - V 3.8gra (Eng) - Rev 2.2 - Round 2.docx

Institut Teknologi Dirgantara Adisutjipto

Document Details

Submission ID
trn:oid:::3618:156238711

Submission Date
Oct 28, 2026, 9:12 PM GMT+7

Download Date
Oct 28, 2026, 9:23 PM GMT+7

File Name
2026-Paper 3 - DSR ITSP - Bambang S - Template IJIES - INASS - V 3.8gra (Eng) - Rev 2.2 - Round 2.docx

File Size
293.2 KB

21 Pages

10,183 Words

62,417 Characters

4% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
- Quoted Text
- Cited Text
- Small Matches (less than 10 words)
- Internet sources

Match Groups

26 Not Cited or Quoted 4%
Matches with neither in-text citation nor quotation marks

0 Missing Quotations 0%
Matches that are still very similar to source material

0 Missing Citation 0%
Matches that have quotation marks, but no in-text citation

0 Cited and Quoted 0%
Matches with in-text citation present, but no quotation marks

Top Sources

0% Internet sources
3% Publications
4% Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

Match Groups

- **26** Not Cited or Quoted 4%
  Matches with neither in-text citation nor quotation marks
- **0** Missing Quotations 0%
  Matches that are still very similar to source material
- **0** Missing Citation 0%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

0% Internet sources
3% Publications
4% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Submitted works	Jawaharlal Nehru Technological University Kakinada on 2026-08-21 — 3%
2	Submitted works	University of South Africa on 2026-09-28 — <1%
3	Publication	"Dynamics of Hybrid Collaborative Networks", Springer Science and Business Me... — <1%
4	Publication	Sherri Douville. "Advanced Health Technology - Managing Risk While Tackling Bar... — <1%
5	Submitted works	Barry University on 2021-04-19 — <1%
6	Submitted works	TURNITIN UDEA on 2026-04-12 — <1%
7	Submitted works	City University Malaysia on 2026-08-10 — <1%
8	Submitted works	Faculty of Computer Science & Information Technology on 2026-06-01 — <1%

Received: October 18, 2021. Revised: December 20, 2021.1

INASS

International Journal of
Intelligent Engineering & Systems

http://www.inass.org/

A Design Science Research Approach for Developing an Integrated Information Technology Strategic Planning Framework

Bambang Suhartono 1,2; Tole Sutikno 1; Imam Riadi 1

1 Doctoral Program in Informatics, Universitas Ahmad Dahlan, Yogyakarta, Indonesia
2 Informatics Engineering, Universitas Insan Pembangunan Indonesia, Banten, Indonesia
* Corresponding author's Email: bambangsuhartono46@gmail.com , 2437083015@webmail.uad.ac.id

Abstract: Information technology strategic planning is required to align digital investment, enterprise architecture, and IT governance with organizational objectives. However, strategic analysis, architecture design, and governance controls are often addressed separately. This research proposes PRASASTI, an integrated framework combining Ward & Peppard, TOGAF ADM, and COBIT 2019 through a Design Science Research (DSR) approach. The framework connects business and IS/IT analysis, application portfolio planning, enterprise architecture, governance objectives, capability assessment, risk control, an expert-assessed applicability index, and performance measurement. Validation used expert judgment and a Delphi process involving 19 respondents. Across eight framework dimensions, the mean expert-assessed applicability score was 71%, compared with the study-defined evaluation benchmark of 85%, yielding an average gap of 14 percentage points. The dimension-level gaps were 7 percentage points for strategic alignment, 10 for IS/IT portfolio planning, 13 for enterprise architecture blueprint, 16 for governance objective mapping, 19 for risk and control integration, 15 for capability assessment, 17 for performance measurement, and 12 for implementation roadmap. The mean score represented approximately 84% of the 85% benchmark; this ratio is distinct from the 71% applicability score. A quantitative comparison using respondent-level ratings across nine common criteria yielded normalized scores of 54% for Ward & Peppard, 61% for TOGAF ADM, 66% for COBIT 2019, and 81% for PRASASTI. Paired-samples tests indicated higher PRASASTI ratings in this sample after Bonferroni correction. These findings represent expert assessments of framework artifacts rather than evidence of improved organizational outcomes. PRASASTI is proposed as a structured basis for IT Master Plan development and digital transformation planning.

Commented [Bambang1]: Reviewer 1 - Comment 8 (has updated/ revised)

.

Keywords: Information technology strategic planning, Ward & Peppard, TOGAF, COBIT 2019, enterprise architecture, IT governance , Design Science Research

Received: [Date]. Revised: [Date].

1. Introduction

Information technology strategic planning (IT) has become an important activity for organizations that want to ensure that IT investments support the organization's objectives and generate sustainable income benefits [1]. Digital transformation and the acceleration of new technology adoption will require a structured, capable plan to handle complexity across functions [2]. On the other hand, many organizations still experience misalignment between business strategy and IT initiatives, which results in suboptimal investment outcomes [3]. IT Planning strategy is a crucial element in ensuring harmony between the business organization's objectives and the use of information technology, especially amid

This article is licensed under a Creative Commons Attribution- ShareAlike 4.0 International License.

Received: October 18, 2021. Revised: December 20, 2021.2

complex digital disruption [4]. In practice, many organizations still face challenges in compiling structured, measurable, and effective IT plans that comprehensively meet business needs. Frequent problems that happen are the existence gap between business strategy and IT strategy, the weakness mapping needs application , not yet optimal IT governance, as well as no integration between strategic planning, enterprise architecture , and control mechanisms. These conditions cause IT initiatives to often run partially, with no good, and not yet a fully value-oriented business. A successful IT planning strategy is heavily influenced by the alignment of business and IT strategies, management support, effective IT governance, and organizational capability [4].

.

Previous studies show that various frameworks have been used in IT strategic planning. Ward & Peppard are often used to analyze business conditions and systems information , formulate IT strategy needs, and compile portfolio applications. TOGAF ADM is used to build enterprise architecture, including business architecture, data, applications, and technology. While that , COBIT 2019 is used to ensure IT governance and management is running in a way structured through governance objectives , control , measurement of performance , as well as Risk management [5]. In this systematic study of 62 articles from 2015-2024 [4], the Ward & Peppard framework was the most widely used in PSTI, appearing in 29 of the 62 articles (around 47%), followed by TOGAF in 15 of the 62 articles (around 24%). Some frameworks have been widely used to support IT strategy planning. Ward & Peppard present an approach to planning strategic IS/IT that emphasizes analysis of the business environment, formulation of needs, and planning the application portfolio, so that IT planning can be aligned with business objectives [6]. Approach the Lots used in case studies; the latest to implement a combination of SWOT analysis, PEST, CSF, McFarlan grid, and business mapping capability to compile an IS/IT plan [6] [7]. While doing so, TOGAF provides a comprehensive Architecture Development Method (ADM) for designing enterprise architecture (EA) — including business architecture, data, applications, and technology — as well as artifacts (catalogs, matrices, diagrams) that help translate business strategy into a technical blueprint [8]. Empirical studies of more cases show that TOGAF-based EA practices can accelerate organizational transformation when combined with measurement benefits and strong governance [9] [3].

On the governance side, COBIT 2019 sets out comprehensive objectives for governance & management. To ensure value , control risk, and resilience of IT resources [10]. COBIT 2019 is designed to integrate easily with other frameworks (e.g., TOGAF, ITIL, ISO), thereby contextualizing EA for practical governance [10]. Literature review: The latest literature also highlights the role of COBIT in aligning IT strategy with management risk, measuring performance, and governance source power — the essential thing when an organization is implementing an enterprise architecture roadmap [11].

While each framework has strengths in specific domains, partial implementations of the frameworks still face various limitations. Ward & Peppard is particularly strong in the strategic aspect. Perspective, but it has not explicitly provided an enterprise development mechanism, architecture, and governance. TOGAF ADM excels in enterprise architecture development but places relatively little emphasis on governance mechanisms, monitoring, and comprehensive IT performance evaluation. On the other hand, COBIT 2019 has a strong orientation toward governance and management information, as well as toward technology companies. Although COBIT 2019 includes APO02 *Managed Strategy* and APO03 *Managed Enterprise Architecture* , this framework does not provide a detailed and comprehensive methodology for formulating business strategies from analysis of the environment until determining portfolio priorities, or for developing *baseline architecture* , *target architecture* , analyzing gaps , and mapping the road migration as provided by the strategic planning methodology and TOGAF ADM [10] [12] [8] .

The state of the art shows increasing attention to integrating strategic planning, enterprise architecture, and IT governance to support an organization's digital transformation. Several studies have attempted to combine enterprise architecture with a governance framework to improve the effectiveness of IT implementation and organizational change management [13][14][15][16]. The literature shows that enterprise architecture and IT governance play complementary roles, but integrating them into a single validated operation remains limited [17][18][19][20].

Based on these conditions, a research gap remains: the lack of an integrated conceptual framework that simultaneously incorporates Ward & Peppard, TOGAF ADM, and COBIT 2019 into a single adaptive strategic IT planning model. Research has previously discussed the planning framework, strategy, enterprise architecture, and IT

This article is licensed under a Creative Commons Attribution- ShareAlike 4.0 International License.

Received: October 18, 2021. Revised: December 20, 2021.3

governance separately or in partial combination. The study's IT planning strategy identified Ward and Peppard, TOGAF, and COBIT as relevant approaches but has not yet demonstrated a mechanism to orchestrate all three into a single integrated model [4]. These limitations have the potential to hinder the maintenance of business–IT alignment, digital transformation coherence, and systematic, evidence-based IT performance measurement [21][22][23].

To address this gap, this study proposes a conceptual framework called PRASASTI (Integrated Strategic IT Planning Framework), a hybrid model that integrates Ward & Peppard, TOGAF ADM, and COBIT 2019 into a single, unified IT strategic planning framework. In the proposed framework , Ward & Peppard are used as the strategic perspective to analyze business needs and organizational priorities, TOGAF ADM is used as an architectural perspective for building an enterprise architecture systematically, while COBIT 2019 is used as a governance perspective to ensure monitoring , evaluation, control, and realization of IT benefits run continuously.

The main contribution of this research is the development of a multidimensional conceptual model that links the domains of strategy, enterprise architecture, and IT governance into a single, integrated cycle. strategic IT planning . This research also produces a mapping of the relationships among frameworks to clarify each domain's contribution to supporting strategic IT planning. organizational alignment . Furthermore, this strategic planning framework serves as an adaptive conceptual foundation for developing and implementing IT master plans in modern organizations. This study uses a design science research approach because its main objective is to develop conceptual artifacts in the form of a framework. [24] Explain that design science research is relevant for studying system information because it is oriented towards creating and evaluating artifacts that can give marked practical and scientific value.

The remainder of this paper is organized as follows. Section 2 presents the theoretical background related to IT strategic planning, enterprise architecture, and IT governance frameworks. Section 3 discusses the research gap and novelty of the proposed PRASASTI framework. Section 4 describes the Design Science Research methodology used for artifact development and validation. Section 5 presents the proposed framework and its integration mechanism. Section 6 discusses the validation results, expert assessment, and gap analysis. Finally, Section 7 presents the discussion, research limitations, and conclusions.this

2. Literature Review

2.1. Planning IT Strategy

IT strategy planning is a systematic process for identifying business needs, evaluating current IS/IT conditions, determining priority applications, designing infrastructure, and preparing an implementation roadmap. In practice, planning an IT strategy must answer questions about the system the organization needs, how the system supports business processes, what technology is required, and how IT investments deliver value to the organization.

[25] Explain that planning strategic system information is required when system information utilization is not yet at maximum capacity, as the document planning strategy does not yet exist and will serve as a reference for development. This strengthens the argument that planning an IT strategy is not merely a list of projects, but must be based on an analysis of the organization's needs and priorities.

2.2. Ward & Peppard

Ward & Peppard emphasize that the IS/IT strategy must start with a business approach that covers analysis of the internal and external business environments, as well as the internal and external IS/IT environments. The main outputs include system strategy information, technology strategy information, IS/IT management strategy, and future portfolio applications.

Study [26] applies Ward & Peppard to e-commerce companies and shows that this approach can produce a recommendation portfolio application. The study confirms that SWOT analysis, Value Chain, PEST, CSF, and McFarlan Strategic Grid are important tools for strategic planning in IS/IT. According to [27], Ward & Peppard can help organizations that do not have their own integrated systems formulate the need for IS/IT development applications and strategies.

Ward & Peppard's limitations lie in their not yet having provided detailed mechanisms for translating the IS/IT strategy into a complete enterprise architecture. Therefore, this framework needs to be equipped with enterprise architecture approaches such as TOGAF.

This article is licensed under a Creative Commons Attribution- ShareAlike 4.0 International License.

Received: October 18, 2021. Revised: December 20, 2021.4

2.3. TOGAF and Enterprise Architecture

Enterprise architecture is an approach to describing the connections among business processes, data, applications, technology, and infrastructure within an organization. TOGAF provides an Architecture Development Method for compiling the architecture vision, business architecture, data architecture, application architecture, technology architecture, opportunities and solutions, migration planning, implementation governance, and architecture change management.

According to [28] enterprise architecture can enable digital transformation by helping organizations align IT strategy, technology, processes, and business objectives. Meanwhile, according to [29], TOGAF 10 can be used to support digital transformation in organizations, especially small and medium-sized ones, through more enterprise architecture design.

Although TOGAF is strong in architectural blueprint preparation , it does not yet approach this in a way that specifically provides mechanism analysis for portfolio application in business strategy, as proposed by Ward & Peppard. In addition, TOGAF does not provide capability assessment and governance objectives in a comprehensive, explicit manner, as COBIT 2019

2.4. COBIT 2019 and IT Governance

COBIT 2019 is an IT governance and management framework that is oriented towards creating value , managing risk , optimizing resource power , and measuring performance . COBIT 2019 domains include: Evaluate, Direct, and Monitor; Align, Plan, and Organize; Build, Acquire, and Implement; Deliver, Service, and Support; and Monitor, Evaluate, and Assess.

[30] Using COBIT 2019 to assess the capability level of the system information library . Research findings indicate that COBIT 2019 can help assess the current state, set targets, and develop recommendations to improve IT governance. [31] confirms that the COBIT 2019 Design Factor Toolkit can help map governance objectives and produce gap scores as a basis for IT governance improvements .

COBIT 2019's strengths lie in its governance objectives, capability assessment, risk control, and performance measurement. However, COBIT 2019 is not a primary framework. For a comprehensive analysis of strategic IS/IT and enterprise architecture blueprint. Therefore, COBIT 2019 is necessarily integrated with Ward & Peppard and TOGAF.

2.5. Framework Integration

A number of recent studies have combined the two frameworks, but few have yet integrated Ward & Peppard, TOGAF, and COBIT 2019 into a single end-to-end model. [32] Adaptation planning, strategic, and information systems with COBIT 2019 at universities, post-COVID-19. Research shows that COBIT 2019 can be used to determine governance domains and to conduct capability gap analysis after the strategic planning process is complete.

[33] Integrating COBIT 2019 and TOGAF ADM for designing enterprise architecture based on risk technology information . The integration is relevant because it can connect IT risks to enterprise architecture design. However, the approach is not yet explicitly included in Ward & Peppard as a baseline analysis for strategic IS/IT and portfolio applications.

Therefore, this research fills the gap by integrating three frameworks: Ward & Peppard as the basis for strategic IS/IT analysis, TOGAF as the basis for preparing the enterprise architecture blueprint, and COBIT 2019 as the governance principles, capabilities, risks, controls, and performance measurement..

Table 1. Summary study previously

Reference	Summary findings and gaps
Andry et al. (2023)	Ward & Peppard produces portfolio application for e-commerce; not yet covers enterprise architecture and governance.
Desyana & Sitokdana (2022)	Ward & Peppard helps identification IS/IT based needs analysis business ; not yet measure governance capabilities .
Harits et al. (2022)	Ward & Peppard and COBIT 2019 support capability gap analysis; not yet generate TOGAF blueprints.
Rahimi et al. (2023)	Enterprise architecture becomes an enabler of digital transformation ; not yet focus on governance objectives.
Safitri et al. (2024)	TOGAF 10 supports SME digital transformation ; not yet integrate COBIT 2019 full .
Christian et al. (2023)	COBIT 2019 assesses capability levels and recommendations repair ; not yet connect portfolio application .

This article is licensed under a Creative Commons Attribution- ShareAlike 4.0 International License.

Received: October 18, 2021. Revised: December 20, 2021.5

Reference	Summary findings and gaps
Mubarak & Fianty (2023)	The COBIT 2019 Design Factor Toolkit determines governance priorities ; not yet building enterprise architecture.
Putra et al. (2025)	TOGAF and COBIT 2019 integrate IT architecture and risks ; not yet enter Ward & Peppard.

To further justify the identified research gaps, a comparative analysis was conducted among existing IT strategic planning frameworks, including Ward and Peppard, TOGAF, COBIT 2019, and previous framework integration approaches. The comparison was performed based on several critical dimensions required in comprehensive IT strategic planning, including strategic analysis, application portfolio management, enterprise architecture development, IT governance, capability assessment, risk management, and performance measurement. This analysis aims to identify the strengths and limitations of each framework and demonstrate the necessity of developing an integrated framework that addresses the gaps between strategic planning, architecture design, governance, and continuous performance evaluation.

Table 2 . Comparative Analysis of Existing IT Strategic Planning Frameworks

Capability Dimension	WP	TOG	COB	PRA
Business strategy analysis	S	P	P	I
IT requirement identification	S	P	L	I
Enterprise architecture blueprint	L	S	L	I
Application portfolio prioritization	S	P	P	I
Governance objectives	L	P	S	I
Capability assessment	L	P	S	I
Risk and control	L	P	S	I
Performance measurement	L	P	S	I
Requirement-to-control traceability	X	P	P	PM
Priority propagation mechanism	X	X	X	PM
Cross-framework feedback loop	X	P	P	PM

Noted :
WP=Ward & Peppard;
TOG=TOGAF;COB=COBIT 2019; PRA=Prasasti

S = Strong ; P=Partial; L=Limited; I=Integrated; PM=Proposed Mechanisme ; X=not exist

The table no. 2, presents a comparative analysis of capability dimensions among Ward and Peppard, TOGAF ADM, COBIT 2019, and PRASASTI. The comparison shows that each existing framework provides strong capabilities in specific areas: Ward and Peppard focuses on strategic analysis and IT portfolio planning, TOGAF ADM emphasizes enterprise architecture development, while COBIT 2019 provides governance, control, capability assessment, and performance management.
However, the comparison highlights several integration gaps in existing frameworks, particularly in requirement-to-control traceability, priority propagation, and cross-framework feedback mechanisms. PRASASTI addresses these limitations by introducing an integrated mechanism that connects strategic requirements, enterprise architecture decisions, governance objectives, and continuous improvement processes. Therefore, PRASASTI extends beyond framework combination by providing structured integration and traceability across IT strategic planning domains.

3. Research Gap and Novelty

Based on the library review, there are three major research gaps. First , the integration of Ward & Peppard and TOGAF is strong. In IS/IT strategy analysis and enterprise architecture, it does not yet explicitly provide governance objectives, capability assessment, risk control, and performance measurement. Second, the integration of Ward & Peppard and COBIT 2019 is strong in IS/IT strategy and governance, but has not yet produced a complete enterprise architecture blueprint. Third, the integration of TOGAF and COBIT 2019 is strong in architecture and governance, but not yet strong enough in strategic analysis of IS/IT and in compiling an application portfolio aligned with business strategy.

Research novelty: This lies in the development of an integrated framework that combines Ward & Peppard, TOGAF, and COBIT 2019 into a single planning model for IT strategy. This framework connects business strategy analysis, portfolio application, enterprise architecture blueprint, governance objectives, capability assessment, risk control, readiness index, gap analysis, and performance measurement. Additionally, the

This article is licensed under a Creative Commons Attribution- ShareAlike 4.0 International License.

Received: October 18, 2021. Revised: December 20, 2021.6

research uses the Design Science Research approach [24] to build conceptual artifacts and the Delphi method [34][35] to validate the model through expert judgment.

Moreover, this integrated framework provides a comprehensive approach to IT strategic planning by unifying strategy analysis, enterprise architecture, and governance objectives within a single model. It enhances alignment between business goals and IT initiatives while facilitating effective resource allocation and portfolio prioritization. The framework also incorporates readiness and gap analysis, allowing organizations to identify weaknesses and implement corrective measures proactively. Through performance measurement and capability assessment, the model supports continuous improvement and informed decision-making. The inclusion of expert validation through the Delphi method enhances the framework's reliability and applicability. Overall, this research contributes a practical and theoretically grounded approach for organizations seeking structured, integrated IT strategic planning.



Fig. 1 Conceptual model of the PRASASTI Framework

In the first stage, the organization's vision, mission, and goals form the basis for strategic analysis. Second stage using Ward & Peppard to evaluate the business environment and IS/IT. The third stage produces a portfolio application strategy. Fourth stage: using TOGAF to build architecture for business, data, applications, and technology. Fifth stage mapping COBIT 2019 governance objectives. Sixth stage measuring capability levels, risks and controls . Seventh stage: prepare the implementation roadmap, KPI, readiness index, and gap analysis.

4. Proposed Framework

Proposed framework named PRASASTI Framework (Integrated Strategic, Architecture and Governance Framework for IT Strategic Planning). This model consists of seven main stages: organizational strategy analysis , analysis of strategic IS/IT, formulation of portfolio applications , enterprise architecture blueprint preparation , governance objectives mapping , capability assessment and risk control, as well as roadmap and performance measurement.

Table 3. Components of the PRASASTI Framework

Component	Sources and functions
Strategic analysis	Ward & Peppard: Identifying need business and IS/IT direction
Application portfolio	Ward & Peppard: Defining priority future applications
Business architecture	TOGAF: Mapping business processes and capabilities organization
Data architecture	TOGAF: Designing data structure , flow , and ownership
Application architecture	TOGAF: Defining application targets and integration system
Technology architecture	TOGAF: Designing platforms, networks , infrastructure , and security
Governance objectives	COBIT 2019: Connecting objective organization with IT governance objectives
Capability assessment	COBIT 2019: Measuring current and target capability levels
Risk and control	COBIT 2019: Managing IT risks and controls
Performance measurement	COBIT 2019: Defining KPIs and monitoring roadmap

This article is licensed under a Creative Commons Attribution- ShareAlike 4.0 International License.

Received: October 18, 2021. Revised: December 20, 2021.7

Table 3 above shows that the PRASASTI framework was built by integrating three main approaches: Ward & Peppard, TOGAF, and COBIT 2019. Ward & Peppard is used in the initial component, as a *strategic analysis* and *application portfolio,* to identify business needs and IS/IT direction and to determine future application priorities. Thus, the initial phase of this framework serves as a basis for aligning organizational strategy with the needs of information systems development.

Furthermore, TOGAF is used to strengthen enterprise architecture by integrating *business components. architecture, data architecture, application architecture, and technology.* These four components help organizations map business processes, data structures, application integration, and technology infrastructure needs. Meanwhile, COBIT 2019 complements *the* framework by addressing IT governance, capability measurement, risk management, control, and KPI determination and monitoring. roadmap . The integration of these three frameworks makes PRASASTI a more comprehensive IT strategic planning framework, as it covers strategy, architecture, governance, risk, and performance measurement.

Table 4. Variables, Indicator and references

Variables	Indicators and references
Strategic alignment	Compliance objective business and IT (Ward & Peppard)
Strategic analysis	SWOT, PEST, Value Chain, CSF (Ward & Peppard)
Application portfolio	McFarlan Strategic Grid and priorities application (Ward & Peppard)
Enterprise architecture	Architecture business , data, applications , technology (TOGAF)
Governance objectives	Mapping objective organization to governance objectives (COBIT 2019)
Capability assessment	Current capability and target capability (COBIT 2019)
Risk control	Identification risk , control , and mitigation (COBIT 2019)
Performance measurement	KPI, readiness index, and monitoring (COBIT 2019)

5. Research Methodology

This study uses a design science research approach . This approach was chosen because the study aims to produce conceptual artifacts in the form of a planning framework for IT strategy that can be used to solve practical problems in organizations. Stages of the study covering identification of the problem, determination of the solution, design and development of objectives, artifacts, demonstrations, evaluation, and communication, as shown in the figure below.

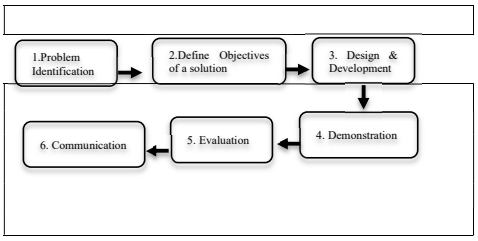


Fig. 2. Stages Methodology Research

Data collection techniques include literature studies, framework analysis, instrument compilation, expert judgment questionnaire, Delphi round 1, Delphi round 2, and documentation comment expert. Instrument use Likert scale (1 to 5) [36] to evaluate the relevance, clarity, completeness, measurability, and feasibility of implementation for every dimension and indicator of the framework.

This study also uses expert judgment with the Delphi method to evaluate and validate the developed results, content validity ratio, and Aiken's V [37][38] . The reliability of an instrument can be assessed using Cronbach's alpha.

5.1. formula Testing

Validity content can be calculated using the content validity ratio and Aiken's V. Reliability of the instrument can be calculated using Cronbach's alpha. Delphi consensus can be analyzed using the interquartile range. In this study, the expert-assessed applicability index is used to measure the perceived completeness, suitability, and implementation feasibility of each PRASASTI framework dimension based on expert evaluation. This index does not represent organizational maturity or actual organizational readiness

As for the formulation for content validity ratio:

$$CVR\ i = (n\ e - N/2) / (N/2) \qquad (1)$$

Information :

This article is licensed under a Creative Commons Attribution- ShareAlike 4.0 International License.

Received: October 18, 2021. Revised: December 20, 2021.8

Respondents' validation was designed for up to 19 people, including expert planning IT strategists, enterprise architecture experts, IT governance experts, IT practitioners, and academic system information experts. Amount: This is used for expert judgment and Delphi,

n_e = number the expert who rated the item as important
N= number of all experts

3. Variables and indicators study

The formulation of Aiken's V is:

$$V = (s) / [n(c - 1)] \quad (2)$$

Description :
s=r−l $_o$

r= score given by expert
l_o= lowest score
n= number of experts
c= number of assessment categories

The Cronbach's Alpha formula is :

$$\alpha = \frac{k}{(k-1)} * \left(1 - \frac{\Sigma\sigma_i^2}{\sigma_i^2}\right) \quad (3)$$

Description :
k= number of items
σ_i^2= item variance
σ_t^2= total variance

As for the formulation applicabiltiy Index:

$$Index = (AS / MS) \times 100\% \quad (4)$$

Description:
Index = normalized applicability or achievement index, expressed as a percentage.
AS= Actual score, score obtained from the assessment or evaluation.
MS = Maximum score , highest possible score under the assessment instrument.
100% = conversion factor used to express the ratio as a percentage.

All statistical indicators, including Content Validity Ratio (CVR), Content Validity Index (CVI), Aiken's V, Cronbach's Alpha, and IQR, were calculated using the original numerical values. For presentation purposes, the results were rounded consistently to two decimal places to improve readability while maintaining calculation accuracy.

Quantitative comparative evaluation and paired-samples t-test

In addition to expert-assessed applicability and content validation, the quantitative comparison uses respondent-level ratings from the QC Data worksheet. The same 19 respondents evaluated Ward & Peppard, TOGAF ADM, COBIT 2019, and PRASASTI using a five-point scale. Criterion means and normalized scores are summarized in Tables 11 and 12; paired comparisons are reported in Table 13. The calculations use each respondent's mean across the nine criteria, preserving the paired design.

Criterion mean scores
For each approach and evaluation criterion, the mean score is calculated by summing the ratings from all respondents and dividing by the number of valid respondents:

$$\bar{x}_j = (\Sigma_{i=1}^{n} x_{ij}) / n \quad (5)$$

Description:
$\bar{x}_j$ = mean score for evaluation criterion j.
x_{ij} = score assigned by respondent i to criterion j.
n = number of valid respondents (19 respondents in this analysis).
i = respondent index, ranging from 1 to n.
j = evaluation-criterion index, ranging from 1 to 9.
Σ = summation over all respondents.
Here, x_{ij} is the rating given by respondent i for criterion j, and $\bar{x}_j$ is the mean rating for that criterion. The mean scores are reported on the original 1–5 scale and summarized in Table 9. The nine criteria describe complementary capabilities; therefore, the criterion-level results should also be interpreted individually rather than relying only on the aggregate score.

Aggregate and normalized scores
The aggregate score for each approach is the sum of the nine criterion means. Because there are nine criteria and the maximum rating for each criterion is 5, the maximum possible aggregate score is:

$$S_{max} = k \times R_{max} \quad (6)$$

Description:
S_{max} = maximum possible aggregate score across all evaluation criteria.
k = number of evaluation criteria
R_{max} = maximum rating assigned to each criterion (5 points).

This article is licensed under a Creative Commons Attribution- ShareAlike 4.0 International License.

Received: October 18, 2021. Revised: December 20, 2021.9

$$NS\ (\%) = (S / S_{max}) \times 100 \qquad (7)$$

Description:
NS = normalized score, expressed as a percentage.
S = aggregate score, calculated as the sum of the nine criterion mean scores.
S_{max} = maximum possible aggregate score
$\bar{x}_j$ = mean score for evaluation criterion j.
j = evaluation-criterion index, ranging from 1 to 9.
100 = conversion factor for expressing the normalized score as a percentage.

Paired-samples t-test
A paired-samples t-test was used because the same respondents rated PRASASTI and each baseline approach. For each respondent i, the paired difference is calculated as:

$$d_i = P_i - C_i \qquad (8)$$

Description:
P_i = respondent i's PRASASTI score and
C_i = Respondent's score for the comparator (Ward & Peppard, TOGAF ADM, or COBIT 2019).

The mean and standard deviation of the paired differences are:

$$\bar{d} = (\Sigma_{i=1}^{n} d_i) / n; \quad s_d = \sqrt{[\Sigma_{i=1}^{n}(d_i - \bar{d})^2 / (n - 1)]}. \qquad (9)$$

Description:
$\bar{d}$ = mean of the paired differences.
d_i = score difference for respondent i, calculated as the PRASASTI score minus the comparator score.
s_d = sample standard deviation of the paired differences.
n = number of paired observations (19 respondents).
i = respondent index, ranging from 1 to n.
Σ = summation over all paired observations.

Operational decision rules for applying PRASASTI

The following decision rules make the framework's workflow more explicit. First, document the organizational objectives, scope, constraints, stakeholders, and available evidence before analysis. Second, use Ward & Peppard techniques (including business/IS-IT analysis, SWOT, PEST, CSF, and Value Chain where relevant) to record business drivers and information-system needs; each selected technique must be linked to a stated question or decision. Third, include an application in the McFarlan Grid only when its strategic and operational roles can be justified from documented evidence, and record the rationale for its portfolio classification. Fourth, select TOGAF ADM phases and architecture deliverables according to the scope and the decisions required; record the input, rationale, owner, and output for every selected phase rather than assuming that all artifacts are mandatory in every case. Fifth, select COBIT 2019 governance objectives and design factors by tracing each one to a documented business need, risk, compliance obligation, or governance requirement. Capability targets and risk-acceptance thresholds must be set and approved by the organization before scoring; this study does not prescribe a universal target for all organizations. Sixth, when priorities conflict, document the competing requirements and compare them against mandatory compliance obligations, risk severity, strategic alignment, dependencies, and feasibility. If the conflict remains unresolved, escalate it to the designated governance/steering authority and record the decision and rationale. Finally, define acceptance criteria and performance indicators before implementation, and revisit earlier analysis when evidence changes or a requirement cannot be traced to an approved objective. These rules are proposed operational guidance and require validation through case-based implementation studies.

6. Results

As for the results discussion in study This is as following :

6.1 Problem Identification

The first stage is problem identification. At this stage, the researcher analyzes the main problem that occurred in the object of research. In the study of the development of a planning framework, the main IT strategic issue identified is that many organizations do not yet have an integrated IT planning framework. IT strategy planning is often still done in a partial way, for example, by only compiling a list of applications, only creating architectural blueprints, or only conducting an IT governance audit without connecting it to the business strategy and organization.

At this stage, the researcher finds that Ward and Peppard's approach is strong in analyzing strategic IS/IT and portfolio applications, but does not yet provide a detailed design of enterprise architecture in general. TOGAF is strong in compilation architecture, business , data, applications, and

This article is licensed under a Creative Commons Attribution- ShareAlike 4.0 International License.

Received: October 18, 2021. Revised: December 20, 2021.10

technology, but not yet in ways that provide specialized strategic analysis and application portfolio approaches, such as those of Ward and Peppard. Meanwhile, COBIT 2019 is strong in IT governance, capability assessment, risk control, and performance measurement, but not the main framework for developing IS/IT strategies and enterprise architecture blueprints. Therefore, the main problem is not yet the lack of a framework that integrates aspects of strategy, architecture, governance, risk, and performance measurement.

6.2 Define objectives of a solution

The stage two is to define the objective of the solution to be developed. The goal of the solution is to develop a new framework that integrates three main frameworks: Ward and Peppard, TOGAF, and COBIT 2019.

Solution goals that can be explained in terms of a number of targets. First, the framework must be capable of helping organizations conduct strategic business analysis and IS/IT analysis. Second , the framework must be capable of producing a portfolio-appropriate application for the organization's needs. Third , the framework must be capable of translating IS/IT strategy into an enterprise architecture blueprint. Fourth , the framework must provide a mechanism for mapping governance objectives and capability assessment. Fifth, the framework must include aspects of risk control, performance measurement, a readiness index, and gap analysis.

6.3 Design and development

The third stage is design and development, namely the process of developing artifacts and research . This framework is the result of integrating three main sources: Ward and Peppard for strategic analysis and application portfolio aspects; TOGAF for enterprise architecture aspects; and COBIT 2019 for aspects of governance, capability, risk, control, and performance measurement. At this stage, the researcher begins compiling the framework structure. Ward and Peppard's components are used in the initial framework for identifying the need for business, IS/IT direction, internal and external conditions of the organization, as well as priority future applications. TOGAF components are used to compile business, data, application, and technology architecture. COBIT 2019 components are used to map IT governance objectives, measure process capability, manage risk, determine controls, and compile KPIs.

The stages of design and development of the framework artifact can be depicted in Fig. 3 below :



Fig. 3. Artifact Framework Flowchart

As for identification component from each framework can depicted in form table under This :

Table 5. Mapping Framework Components in the Design and Development Stage

No	Source Framework	Main Components	Function in the New Framework	Output Generated
1	Ward and Peppard	Strategic analysis	Identifying need business , direction strategic organization , opportunities , threats , strengths and weaknesses related to IS/IT	Analysis need business and IS/IT strategy direction
2	Ward and Peppard	Internal business analysis	Analyze business processes , resources power , structure organization , and needs internal information	Business process and requirements map internal information
3	Ward and Peppard	External business analysis	Analyze factor external like regulation , competition , technology , and stakeholder needs	Analysis environment external organization
4	Ward and Peppard	Internal IS/IT analysis	Evaluate condition applications , infrastructure , data, IT human resources, and IT services that are currently being walk	Condition existing IS/IT

This article is licensed under a Creative Commons Attribution- ShareAlike 4.0 International License.

Received: October 18, 2021. Revised: December 20, 2021.11

No	Source Framework	Main Components	Function in the New Framework	Output Generated
5	Ward and Peppard	External IS/IT analysis	Identifying trend technology and practice best relevant with organization	Opportunity adoption technology new
6	Ward and Peppard	Application portfolio	Determine priority future applications based on mark strategic and needs organization	Portfolio future applications
7	TOGAF	Architectural vision	Developing a vision harmonious architecture with objective business and IT strategy	Enterprise architecture vision
8	TOGAF	Business architecture	Mapping business processes , functions organizations , services , actors , and capabilities business	Architectural blueprint business
9	TOGAF	Data architecture	Designing data structures , data requirements , data flows , and integration information	Data architecture blueprint
10	TOGAF	Application architecture	Designing connection between applications , integration systems and support application to business processes	Architectural blueprint application
11	TOGAF	Technology architecture	Designing need infrastructure , network , platform, security , and technology supporters	Architectural blueprint technology
12	TOGAF	Migration planning	Determine stages migration from condition moment This going to target condition	Migration roadmap architecture
13	COBIT 2019	Governance objectives	Map objective organization to IT governance objectives	Governance objectives map
14	COBIT 2019	Capability assessment	Measure current IT process capabilities this and determine capability targets	Current capability and target capability values
15	COBIT 2019	Risk and control	Identifying IT risks , controls , and mitigations required	IT risk , control and mitigation register
16	COBIT 2019	Performance measurement	Determine KPIs, monitoring indicators , and measures success roadmap implementation	KPIs and indicators IT performance

As for mapping and similarities and roles from each framework can depicted in form table under This :

Table 6. Mapping similarities and roles from each framework at the design and development stage

No	Integration Aspects	Ward and Peppard	TOGAF	COBIT 2019	Roles in the New Framework
1	Orientation main	Planning strategic IS/IT	Enterprise Architecture	IT governance and management	Combining IT strategy, architecture , and governance in one integrated framework
2	Alignment business and IT	Aligning business strategy with IS/IT strategy	Aligning architecture business , data, applications , and technology	Aligning objective organization with governance objectives	Ensure all over IT initiatives support objective organization
3	Analysis condition organization	Analyze environment internal- external business and IS/IT	Identifying baseline architecture and target architecture	Using enterprise goals and design factors	Become base for understand condition moment this and the need change
4	Planning application	Building a portfolio application moment this and the future	Designing application architecture	Arrange control and management application	Determine priority aligned applications with architecture and governance
5	Design architecture	Not designing architecture in a way details	Developing business, data, application, and technology architecture	Supporting governance of IT architecture and services	TOGAF becomes basic enterprise architecture blueprint

This article is licensed under a Creative Commons Attribution- ShareAlike 4.0 International License.

Received: October 18, 2021. Revised: December 20, 2021.12

No	Integration Aspects	Ward and Peppard	TOGAF	COBIT 2019	Roles in the New Framework
6	IT governance	Developing IS/IT management strategies	Supporting architecture governance	Providing governance and management objectives	COBIT 2019 becomes base IT control , measurement , and governance
7	Measurement capability	Does not provide a detailed process capability model details	Support evaluation readiness architecture	Provide capability assessment	Measure condition moment this , targets, and IT capability gaps
8	Risk and control	Identifying risk strategic through analysis environment	Consider risk in design architecture	Provide risk management and control objectives	Ensuring IT strategy and architecture have control and mitigation risk
9	Implementation roadmap	Produce IS/IT strategic direction and priorities application	Providing migration planning	Provide KPIs and monitoring mechanisms	Produce a targeted and measurable IT implementation roadmap
10	Main output	IS/IT strategy and portfolio application	Enterprise architecture blueprint	Governance objectives, capability level, risk control, and KPI	Produce IT Master Plan, digital transformation roadmap , EA blueprint, and IT governance model

Result of stage This can in the form of table integration framework, framework flowchart , conceptual model , indicators measurement , instruments validation , as well as formula calculations used . As seen in table under this related with mapping integration based on order framework development .

Table 7. Integration Mapping of Ward and Peppard, TOGAF, and COBIT 2019 into the Proposed PRASASTI Framework

No	IT Strategic Planning Stage	Ward & Peppard Contribution	TOGAF ADM Contribution	COBIT 2019 Contribution	Integration Mechanism / Transformation	Output Artifact	Dimension
1	Initiation and Preparation	Identify initial business needs and strategic objectives	Preliminary Phase: define architecture principles, scope, and stakeholders	EDM01, APO01: governance framework and management objectives	Establish governance structure, scope, principles, and planning mechanism for IT Masterplan development	Project Charter, Scope, Governance Structure	Strategic Alignment
2	Business Strategy and Requirement Analysis	Business analysis, internal/external analysis, CSF, SWOT, Value Chain	Phase A Architecture Vision: stakeholder needs and enterprise vision	EDM02, EDM05, APO02: ensure business and IT alignment	Transform business objectives into strategic IT requirements	Business Requirement Map, IT Strategic Requirement	Strategic Alignment

This article is licensed under a Creative Commons Attribution- ShareAlike 4.0 International License.

Received: October 18, 2021. Revised: December 20, 2021.13

No	IT Strategic Planning Stage	Ward & Peppard Contribution	TOGAF ADM Contribution	COBIT 2019 Contribution	Integration Mechanism / Transformation	Output Artifact	Dimension
3	Current SI/IT Assessment	Internal IS/IT analysis, application portfolio, technology assessment	Baseline Architecture (Current State Architecture)	APO03, APO14, BAI09, BAI10, DSS01: assessment of assets, data, and services	Integrate current application, data, technology, and capability conditions	Current State Architecture (As-Is)	IS/IT Portfolio Planning
4	Target Identification	Future IS/IT direction and strategic options	Target Architecture Definition	COBIT Design Factors, Enterprise Goals, Alignment Goals	Determine target conditions based on standards, best practices, and organizational needs	Target Capability and Target Architecture	Enterprise Architecture Blueprint
5	Gap Analysis	Strategic gap analysis between current condition and business requirements	Architecture Gap Analysis	APO11, APO12, MEA01: capability, risk, and performance gaps	Compare As-Is and To-Be conditions to determine improvement priorities	Gap Analysis Report	Capability Assessment and Risk Integration
6	IT Strategy Formulation	IS Strategy, IT Strategy, IS/IT Management Strategy	Architecture Vision and Requirement Management	APO02, APO05, EDM02: strategic alignment and value delivery	Develop IT strategy aligned with organizational strategy	IT Strategy Document	Strategic Alignment
7	Enterprise Architecture Design	Identify application and strategic information requirements	Phase B: Business Architecture; Phase C: Data and Application Architecture; Phase D: Technology Architecture	APO03, APO14, BAI03	Transform IT requirements into target architecture blueprint	Enterprise Architecture Blueprint	Enterprise Architecture Blueprint
8	IT Portfolio and Priority Planning	McFarlan Strategic Grid, application portfolio, investment priority	Phase E: Opportunities and Solutions	APO05, BAI01, BAI02	Determine application and program priorities based on business value, risk, and architecture requirements	IT Portfolio and Investment Priority	IS/IT Portfolio Planning

This article is licensed under a Creative Commons Attribution- ShareAlike 4.0 International License.

Received: October 18, 2021. Revised: December 20, 2021.14

No	IT Strategic Planning Stage	Ward & Peppard Contribution	TOGAF ADM Contribution	COBIT 2019 Contribution	Integration Mechanism / Transformation	Output Artifact	Dimension
9	IT Governance Development	IS/IT Management Strategy	Architecture Governance	EDM01, EDM03, EDM04, APO01, APO13	Integrate governance structure, risk management, controls, and decision rights	IT Governance Framework	Governance Objective Mapping
10	Implementation Roadmap Development	Strategic initiatives and implementation planning	Phase E: Opportunities and Solutions; Phase F: Migration Planning; Phase G: Implementation Governance	BAI01, BAI11, APO05	Develop implementation stages based on priority, dependency, and organizational readiness	IT Implementation Roadmap	Performance Measurement and Implementation Planning
11	Investment and KPI Development	Strategic benefits and investment planning	Architecture Capability Management	MEA01, MEA04, APO06	Link IT investment with business benefits and performance indicators	IT Investment Plan and KPI Framework	Performance Measurement
12	Validation and Finalization	Strategic review and alignment evaluation	Architecture Governance Review	MEA01, MEA03: monitoring, evaluation, and compliance	Validate all artifacts to ensure alignment between strategy, architecture, and governance	Final IT Masterplan Document	Governance and Continuous Improvement

The table above shows that each framework occupies a complementary position. Ward and Peppard are used as the basis for strategic IS/IT analysis, primarily to identify business needs, IS/IT conditions, IS/IT strategies, and application portfolios. TOGAF is used to translate the results of this strategic analysis into enterprise architecture designs through the ADM phases, particularly Architecture Vision, Business Architecture, Data Architecture, and Application Architecture. Architecture , Technology Architecture , Opportunities and Solutions , Migration Planning and Implementation Governance . Meanwhile, COBIT 2019 is used to strengthen governance, capability, risk, control, and performance measurement aspects through EDM, APO, BAI, DSS, and MEA processes.

With this mapping, the new framework not only produces an IS/IT strategy but also provides an architectural blueprint, an implementation roadmap, and governance. objectives, capability assessment, risk control, KPI, and monitoring mechanisms. Methodologically, Ward and Peppard serve as the strategic foundation, TOGAF as the architectural foundation, and COBIT 2019 as the governance and measurement foundation

6.4. Demonstration

The demonstration stage was carried out using an expert-based framework walkthrough approach, employing an expert judgment mechanism and the Delphi method. At this stage, the developed framework artifact was demonstrated to the experts using a help document, the framework design, a flowchart, a table mapping components, and

This article is licensed under a Creative Commons Attribution- ShareAlike 4.0 International License.

Received: October 18, 2021. Revised: December 20, 2021.15

scenarios for using the framework. Experts were requested to examine how the framework can be used in the planning process of IT strategy , starting from strategic analysis .

Instrument questionnaire used to get an evaluation expert to clarify the concept, completeness of components , consistency of flow , relevance of indicators, and integration with inter-frameworks, as well as potential implementation of the framework in the organization . Assessment conducted across a number of Delphi rounds to reach consensus on the eligibility artifact. Input from experts in every field is used to refine the framework before entering the evaluation stage.

6.5 Evaluation

Stage fifth is Evaluation, namely evaluate artifacts that have been developed and demonstrated . In study this , evaluation can done through expert judgment, Delphi method , validity test content ,

6.6 Communication

The sixth stage is Communication, namely conveying the results of the study to community academics and practitioners through journal articles. In the journal article, stage communication is presented as a compilation of parts: introduction, literature review, research method, proposed framework, results and discussion, conclusions, limitations, and bibliography.

Results section: This section uses the results data collection based on a questionnaire to show the method framework and how to present the results.

Table 8. Profile respondents

Category respondents	Amount	Percentage
Planning expert IT strategy	5	26%
Enterprise architecture expert	5	26%
IT governance or COBIT expert	3	16%
IT Practitioner or IT manager	4	21%
Academics system information	2	11%
Total	19	100%

The respondent percentages in Table 8 are rounded to the nearest whole number for presentation; all statistical calculations use the original numerical values.

Planning expert IT strategists and enterprise architecture experts each numbering 5 people (26%), emphasizing the role they play in evaluating alignment of strategy and enterprise architecture . There are 3 IT governance or COBIT experts (16%), providing perspective on IT governance, controls, and policies. IT practitioners or IT managers, numbering 4 people (21%), provide input on the practical, related implementation of the framework in real organizational environments. The academic system information totals 2 people (11%), adding a corner view of theoretical and scientific validity to the framework.

The validity and reliability results for the assessment instrument are presented in Table 9.

Table 9. Validity and reliability test results

Test	Results	Interpretation
CVR	0.68-1.00	Valid
S-CVI/Ave	0.98	Validity very good content
Aiken's V	0.79-0.94	Agreement expert tall
Cronbach's Alpha	0.89	Reliable
IQR Delphi round 2	<= 1	Consensus achieved

Based on Table 9, the instrument's validity and reliability testing indicate high reliability. CVR values range from 0.68 to 1.00, indicating that most items were deemed valid by the experts. S-CVI/Ave of 0.98 indicates that the validity of the content is a very good instrument . Aiken's V results show high agreement among experts (0.79–0.94), while Cronbach's Alpha of 0.89 indicates adequate reliability. The Delphi round 2 IQR ≤ 1 indicates consensus among experts on the suitability and eligibility of each instrument item.

The expert-assessed applicability index and gap analysis evaluate perceived applicability and implementation feasibility of the PRASASTI framework artifact. They compare expert assessment scores with the study-defined evaluation benchmark; they do not measure the readiness of a specific organization.

Table 10. Expert-assessed applicability index and gap analysis

Dimension	Current score / benchmark	Gap (percentage points) / status relative to benchmark
Strategic alignment	78% / 85%	7 pp / below benchmark
IS/IT portfolio planning	75% / 85%	10 pp / below benchmark

Commented [Bambang2]: Reviewer 1 - Comment 4 (has updated/ revised)

This article is licensed under a Creative Commons Attribution- ShareAlike 4.0 International License.

Received: October 18, 2021. Revised: December 20, 2021.16

Dimension	Current score / benchmark	Gap (percentage points) / status relative to benchmark
Enterprise architecture blueprint	72% / 85%	13 pp / below benchmark
Governance objective mapping	69% / 85%	16 pp / below benchmark
Risk and control integration	66% / 85%	19 pp / below benchmark
Capability assessment	70% / 85%	15 pp / below benchmark
Performance measurement	68% / 85%	17 pp / below benchmark
Implementation roadmap	73% / 85%	12 pp / below benchmark
Average	71% / 85%	14 pp / below benchmark

Note: Gap (pp=percentage points) = evaluation target (%) − current score (%). For example, 85% − 78% = 7 percentage points (7 pp). Percentage points describe the arithmetic difference between two percentage values; they are not the same as a relative percentage gap.

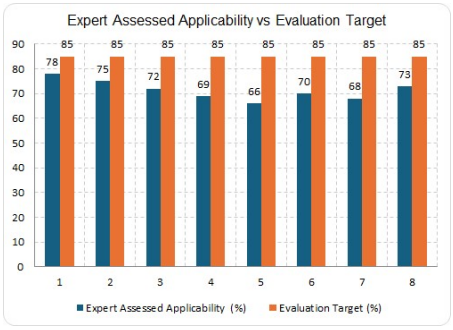


Fig. 4. Expert-assessed applicability scores against the evaluation target.

Table 10 and Fig. 4 show the expert-assessed applicability scores relative to the study-defined 85% evaluation benchmark. The mean across the eight dimensions was 71%, with an average gap of 14 percentage points. Strategic alignment scored 78% (7 pp below the benchmark), whereas risk and control integration scored 66% (19 pp below the benchmark). The other gaps were 10 pp for IS/IT portfolio planning, 13 pp for enterprise architecture blueprint, 16 pp for governance-objective mapping, 15 pp for capability assessment, 17 pp for performance measurement, and 12 pp for the implementation roadmap. The mean score represents approximately 84% of the 85% benchmark. All dimensions remain below the benchmark; no validated organizational maturity category is inferred from these scores.

Quantitative comparative analysis of the four planning approaches

To complement the framework validation, a quantitative comparison used the nine evaluation criteria recorded in the QC Data worksheet. The same 19 respondents rated Ward & Peppard, TOGAF ADM, COBIT 2019, and PRASASTI on a five-point scale. Criterion-level means are reported in Table 11, aggregate normalized scores in Table 12, and paired-samples t-test results for PRASASTI versus each comparator in Table 13.

Table 11. Mean scores by evaluation criterion (scale 1–5)

Evaluation criterion	WP	TOG	COB	PRA
Strategic alignment	3.9	2.9	3.1	3.7
Application portfolio completeness	4.1	3.1	2.5	3.9
Enterprise architecture completeness	2.7	4.4	2.9	4.2
Governance and control coverage	2.5	2.8	4.4	4.2
Risk and capability assessment	2.6	3.2	3.9	4.1
Requirement-to-control traceability	2.0	2.6	3.1	4.0
Priority propagation	1.9	2.7	2.9	4.0
Performance measurement	2.6	3.1	4.1	4.1
Integration and feedback mechanisms	1.9	2.7	2.8	4.2

Commented [Bambang5]: Reviewer 1 - Comment 5,6 (has updated/ revised)

Commented [Bambang3]: Reviewer 1 - Comment 7 (has updated/ revised)

Commented [Bambang6]: Reviewer 1 - Comment 2 (has updated/ revised)

Commented [Bambang4]: Reviewer 1 - Comment 3 (has updated/ revised)

This article is licensed under a Creative Commons Attribution- ShareAlike 4.0 International License.

Received: October 18, 2021. Revised: December 20, 2021.17

Noted :
WP=Ward & Peppard;
TOG=TOGAF;COB=COBIT 2019; PRA=Prasasti

Table 11 shows that the four approaches have different areas of relative strength. Ward & Peppard receives its highest score for application portfolio completeness (4.1) and also scores highly on strategic alignment (3.9). TOGAF ADM has the highest score for enterprise architecture completeness (4.3), consistent with its architecture-oriented role. COBIT 2019 performs strongest in governance and control coverage (4.4), and also scores highly for performance measurement (4.1). PRASASTI scores strongly on integration and feedback mechanisms (4.2), enterprise architecture completeness (4.2), risk and capability assessment (4.1), requirement-to-control traceability (4.0), and priority propagation (4.0). These results suggest that the proposed framework's comparative advantage lies in combining complementary capabilities rather than outperforming the source frameworks on every individual criterion.

Table 12. Aggregate scores and normalized comparison

Approach	Sum of criterion means	Maximum score	Normalized score (%)	Mean score
Ward & Peppard	24.3	45	54	2.7
TOGAF ADM	27.4	45	61	3.0
COBIT 2019	29.7	45	66	3.3
PRASASTI	36.4	45	81	4.0

As shown in Table 12, PRASASTI has the highest aggregate normalized rating (81%), followed by COBIT 2019 (66%), TOGAF ADM (61%), and Ward & Peppard (54%). The normalized score is calculated as the sum of the nine criterion means divided by the maximum possible score of 45, multiplied by 100. It is a descriptive index of ratings under this instrument, not a direct measure of organizational performance or proof of better implementation outcomes.

Table 13. Paired-samples t-test comparing PRASASTI with the baseline approaches

PC	n	Md	SD	t	df	Result
PRA vs WP	19	1.3	0.3	15.5	18	Significant
PRA vs TOG	19	1.0	0.3	13.5	18	Significant
PRA vs COB	19	0.7	0.3	10.2	18	Significant

Noted :
PC=Paired comparisan ; Md=Meandif ; SD=Sddif;
WP=Ward & Peppard;
TOG=TOGAF;COB=COBIT 2019; PRA=Prasasti
The paired-samples t-tests use respondent-level mean scores across the nine criteria (n = 19; df = 18). The Bonferroni-adjusted p-values are below 0.05 for all three comparisons. The confidence intervals are unadjusted 95% intervals; the inferential conclusions are based on the adjusted p-values. The tests assume independent respondent pairs and approximately normally distributed paired differences. Given the small sample, the distribution of differences should be checked; a Wilcoxon signed-rank sensitivity analysis would be useful if this assumption is not tenable.

Sensitivity analysis of the evaluation benchmark

To assess how interpretation changes with the chosen benchmark, the eight dimension scores were compared against alternative thresholds of 80%, 85%, and 90%. The mean observed score remains 71%; the mean gap and the ratio of the mean score to the benchmark change with the threshold. None of the eight dimensions reaches any of these three benchmarks. This is a descriptive sensitivity check, not a validation of the benchmark itself.

Commented [Bambang7]: Reviewer 1 - Comment 2 (has updated/ revised)

Table 14. Sensitivity analysis for alternative evaluation benchmarks

EB	Mean score	Mean gap	Mean score as % of benchmark	Dimensions meeting benchmark
80%	71%	9. pp	89%	0 of 8
85%	71%	14 pp	84%	0 of 8
90%	71%	19 pp	79%	0 of 8

Noted :
EB=Evaluation Benchmark

7. Discussion

Calculation results show that the planning framework integrated the IT strategy's own potential to strengthen the planning IT strategy by linking strategy, architecture, governance, risk, and performance into a single plot. Ward & Peppard provide a base analysis of business and IS/IT. TOGAF translates strategy into an enterprise architecture blueprint, while COBIT 2019 ensures

Commented [Bambang8]: Reviewer 1 - Comment 2 (has updated/ revised)

This article is licensed under a Creative Commons Attribution- ShareAlike 4.0 International License.

Received: October 18, 2021. Revised: December 20, 2021.18

that the design includes governance objectives, capability measurement, risk control, and KPIs.

The biggest gaps in the calculation are in risk and control integration and performance measurement. This shows that organizations often have planned applications and infrastructure but do not yet have control over risks and indicators of strong performance. COBIT 2019 integration is important so that the IT roadmap is not only technical but also measurable, controllable, and sustainably evaluable.

In a theoretical sense, this research expands the draft planning of IT strategy by adding dimensions of enterprise architecture and IT governance. In general, this framework can be used by organizations to develop an IT Master Plan, a digital transformation roadmap, an enterprise architecture blueprint, and an IT governance improvement program.

The quantitative comparison provides complementary evidence from ratings by the same 19 respondents. PRASASTI's respondent-level mean rating was higher than those of Ward & Peppard, TOGAF ADM, and COBIT 2019. Paired-samples t-tests and Bonferroni-adjusted p-values are reported in Table 13. These tests indicate differences in ratings under the defined criteria and sample; they do not establish universal superiority or improved organizational outcomes. The eight-dimension applicability index and the nine-criterion comparative rating are distinct measures and should not be conflated.

7.1 Research Limitations and Future Research Directions

Although PRASASTI has been validated through expert judgment and Delphi evaluation, this study has several limitations. First, evaluation was conducted at the framework-artifact level and does not represent empirical evidence from organizational implementation. Second, the effect of PRASASTI on IT strategic-planning outcomes, decision-making, and organizational performance has not been measured longitudinally. Third, validation focuses on content validity and expert consensus rather than operational performance. Fourth, the 19-person expert sample is limited, and the paired t-test findings depend on the rating instrument and assumptions of the test. Finally, the 85% benchmark is study-defined; the sensitivity analysis shows that conclusions about gaps depend on the chosen threshold.

Data availability and reproducibility. The quantitative comparison is based on the 19 respondent-level records and formulas documented in the study's QC Data and QC_Paired_t_Test worksheets. The calculation workbook should be provided as supplementary material or made available by the corresponding author on reasonable request, subject to participant confidentiality and applicable data-sharing restrictions. The manuscript does not claim that a public repository is currently available. To support independent checking, the shared package should include the questionnaire, codebook, anonymized respondent-level ratings where permitted, calculation formulas, and a README describing scoring, exclusions, rounding, and software used.

8. Conclusion

This study proposes a new integrated model for planning an IT strategic framework, based on Ward & Peppard, TOGAF, and COBIT 2019. This framework was developed using the Design Science Research approach and was designed to be validated through expert judgment and the Delphi method .

Contribution to the main study: This is a development artifact conceptually linking analysis of strategic IS/IT portfolio application , enterprise architecture blueprint, governance objectives, capability assessment, risk control, implementation roadmap , readiness index, gap analysis, and performance measurement. This framework can serve as the basis for a more comprehensive IT Master Plan that is measurable and aligned with the organization's needs.

The study further recommended applying this framework to real-world organizations, such as colleges, high schools, hospitals, cooperative agencies, government agencies, and private-sector companies. Testing is empirically required to evaluate the framework's effectiveness in improving the quality of planning for IT strategy and IT governance.

Conflicts of interest

The authors declare no conflict of interest.

Author contributions

Conceptualization, BS, TS, and IR; methodology, BS; validation, TS, and IR; formal analysis, BS; investigation, B.S.; resources, BS; data curation, BS; writing-original draft preparation, BS; writing-review and editing, TS and IR; visualization, B.S.; supervision, TS and IR.

Acknowledgments

The writer expresses acceptance of the love and support from Universitas Ahmad Dahlan, Yogyakarta, and Universitas Insan Pembangunan

This article is licensed under a Creative Commons Attribution- ShareAlike 4.0 International License.

Received: October 18, 2021. Revised: December 20, 2021.19

Indonesia for the academic support and facilities provided in the compilation of this study.

References

[1] R. van de Wetering, "Dynamic enterprise architecture capabilities and organizational benefits: an empirical mediation study," May 2021, Accessed: Jun. 15, 2026. [Online]. Available: https://arxiv.org/pdf/2105.10036

[2] P. C. Verhoef *et al.*, "Digital transformation: A multidisciplinary reflection and research agenda," *J. Bus. Res.*, vol. 122, no. July 2018, pp. 889–901, 2021, doi: 10.1016/j.jbusres.2019.09.022.

[3] P. Singh, F. Lynch, and M. Helfert, "Enterprise architecture for the transformation of public services based on citizen's feedback," *Digit. Policy, Regul. Gov.*, vol. 26, no. 1, pp. 38–54, Jan. 2024, doi: 10.1108/DPRG-11-2022-0123.

[4] B. Suhartono, T. Sutikno, and ; Imam Riadi, "Decade of IT Strategic Planning : Systematic Review of Frameworks and Critical Success Factors," *JITK (Jurnal Ilmu Pengetah. dan Teknol. Komputer)*, vol. 11, no. 3, pp. 700–714, Feb. 2026, doi: 10.33480/JITK.V11I3.7068.

[5] B. Suhartono, T. Sutikno, and I. Riadi, "Review Framework and Best Practise of COBIT 2019 Standard for IT Governance Implementation," *J. Inf. Syst.*, vol. 12, no. 1, 2025, doi: 10.30656/jsii.v12i1.9926.

[6] P. Merit Biyanti and N. Yalina, "Information System Strategic Planning using Ward and Peppard Method at Universitas Islam Negeri Sunan Ampel Surabaya", doi: 10.5220/0008905800002481.

[7] S. I. Suryana and C. Fibriani, "The IS/IT Strategy Planning Through Ward and Peppard Method in Automotive Spareparts Company," *JINAV J. Inf. Vis.*, vol. 5, no. 1, pp. 1–11, 2024, doi: 10.35877/454ri.jinav2506.

[8] The-Open-Group, *TOGAF® Version 9.1* , vol. Document N. 2011.

[9] V. Bastidas, M. Bezbradica, M. Bilauca, M. Healy, and M. Helfert, "Enterprise Architecture in Smart Cities: Developing an Empirical Grounded Research Agenda," *J. Urban Technol.*, vol. 30, no. 1, pp. 47–70, Jan. 2023, doi: 10.1080/10630732.2022.2122681.

[10] "COBIT®| Control Objectives for Information Technologies®." Accessed: Jun. 17, 2026. [Online]. Available: https://www.isaca.org/resources/cobit

[11] G. M. W. Tangka and E. Lompoliu, "Information {Technology} {Governance} {Using} the {COBIT} 2019 {Framework} at {PT}. {Pelindo} {TPK} {Bitung}," *CogITo Smart J.*, vol. 9, no. 2, pp. 355–367, Dec. 2023, doi: 10.31154/cogito.v9i2.577.355-367.

[12] S. De Haes, W. Van Grembergen, A. Joshi, and T. Huygh, "Enterprise Governance of Information Technology," 2020, doi: 10.1007/978-3-030-25918-1.

[13] Https://www.tandfonline.com/author/Bastidas%2C+Viviana, "Enterprise Architecture in Smart Cities: Developing an Empirical Grounded Research Agenda", [Online]. Available: https://www.tandfonline.com/doi/full/10.1080/10630732.2022.2122681

[14] K. Peffers, T. Tuunanen, M. A. Rothenberger, and S. Chatterjee, "A design science research methodology for information systems research," *J. Manag. Inf. Syst.*, vol. 24, no. 3, pp. 45–77, 2007, doi: 10.2753/MIS0742-1222240302.

[15] T. S. B Suhartono Imam Riadi, "Review Framework dan Best Practise Standar COBIT 2019 untuk Implementasi Tata Kelola TI," *JSiI (Jurnal Sist. Informasi)*, vol. 12, pp. 128–133, 2025, [Online]. Available: https://e-jurnal.lppmunsera.org/index.php/jsii/article/view/9926

[16] I. Riadi, R. Umar, I. Busthomi, and A. W. Muhammad, "Block-hash of blockchain framework against man-in-the-middle attacks," *Regist. J. Ilm. Teknol. Sist. Inf.*, vol. 8, no. 1, pp. 1–9, 2022, doi: 10.26594/register.v8i1.2190.

[17] G. Fuentes-Quijada, F. Ruiz-González, and A. Caro, "Enterprise Architecture and IT Governance to Support the BizDevOps Approach: a Systematic Mapping Study," *Inf. Syst. Front.*, vol. 27, no. 3, pp. 865–888, 2025, doi: 10.1007/s10796-024-10473-2.

[18] A. Wahyudin and Z. A. Hasibuan, "Research

This article is licensed under a Creative Commons Attribution- ShareAlike 4.0 International License.

Received: October 18, 2021. Revised: December 20, 2021.20

classification in strategic information system planning development: A critical review," *Proc. - 2015 Int. Conf. Sci. Inf. Technol. Big Data Spectr. Futur. Inf. Econ. ICSITech 2015*, pp. 287–292, 2016, doi: 10.1109/ICSITech.2015.7407819.

[19] M. Thirasakthana and S. Kiattisin, "Framework De Arquitectura Empresarial Gubernamental Sostenible," *Sustain.*, vol. 13, no. 2, pp. 1–27, 2021.

[20] Sunardi, I. Riadi, and P. A. Raharja, "Vulnerability analysis of E-voting application using open web application security project (OWASP) framework," *Int. J. Adv. Comput. Sci. Appl.*, vol. 10, no. 11, pp. 135–143, 2019, doi: 10.14569/IJACSA.2019.0101118.

[21] B. Anthony Jnr, "Managing digital transformation of smart cities through enterprise architecture–a review and research agenda," *Enterp. Inf. Syst.*, vol. 15, no. 3, pp. 299–331, 2021, doi: 10.1080/17517575.2020.1812006.

[22] A. Hanelt, R. Bohnsack, D. Marz, and C. Antunes Marante, "A Systematic Review of the Literature on Digital Transformation: Insights and Implications for Strategy and Organizational Change," *J. Manag. Stud.*, vol. 58, no. 5, pp. 1159–1197, 2021, doi: 10.1111/joms.12639.

[23] N. R. Busch and A. Zalewski, "A Systematic Literature Review of Enterprise Architecture Evaluation Methods," *ACM Comput. Surv.*, vol. 57, no. 5, 2025, doi: 10.1145/3706582.

[24] K. Peffers, T. Tuunanen, M. A. Rothenberger, and S. Chatterjee, "A design science research methodology for information systems research," *J. Manag. Inf. Syst.*, vol. 24, no. 3, pp. 45–77, Dec. 2007, doi: 10.2753/MIS0742-1222240302.

[25] T. Widayanti and I. Zulkarnaen, "Information Systems Strategic Planning Using the Ward and Peppard Method," *ADI Internaltional Conf. Ser.*, vol. 4, pp. 52–57, 2022, doi: 10.34306/conferenceseries.v4i1.691.

[26] J. F. Andry, D. Y. Bernanda, H. Honni, K. Christianto, and A. Andriani, "Analysis of information systems strategic planning using ward and peppard framework case e-commerce company," *Int. J. Adv. Appl. Sci.*, vol. 12, no. 2, pp. 179–187, Jun. 2023, doi: 10.11591/ijaas.v12.i2.pp179-187.

[27] N. P. Desyana and M. N. N. Sitokdana, "Implementation of Ward and Peppard in Information Systems Strategic Planning (Case Study: Hotel Kartika Bandungan)," *J. Inf. Syst. Informatics*, vol. 4, no. 3, pp. 784–794, Sep. 2022, doi: 10.51519/Journalisi.V4I3.284.

[28] "View of Information Technology Risk-based Enterprise Architecture Design for Training Systems: Integration of COBIT 2019 and TOGAF ADM." Accessed: Jun. 17, 2026. [Online]. Available: https://ejournal.undiksha.ac.id/index.php/JST/article/view/93498/32662

[29] A. A. N. F. Shintya Rahma Safitri1, Rahmat Mulyana2, "View of Developing Enterprise Architecture for BPRACO SMEs Digital Transforamtion by Using TOGAF 10," JIKO (Jurnal Informatika dan Komputer). Accessed: Jun. 01, 2026. [Online]. Available: https://ejournal.unkhair.ac.id/index.php/jiko/article/view/8629/5287

[30] Y. M. Maulana, "Information {Technology} {Governance} {Using} {Control} {Objectives} for {Information} and {Related} {Technology}: Review," *J. Tek. Inform. dan Sist. Inf.*, vol. 9, no. 3, Jan. 2024, doi: 10.28932/jutisi.v9i3.6494.

[31] R. F. Mubarak and M. I. Fianty, "Leveraging COBIT 2019 to Implement IT Governance in Mineral Mining Company," *J. Inf. Syst. Informatics*, vol. 5, no. 3, pp. 1058–1071, Sep. 2023, doi: 10.51519/journalisi.v5i3.545.

[32] A. Harits, R. Gernowo, and D. E. Suseno, "Adaptation of Information Systems Strategic Planning of Universities Using COBIT 2019 in Post Covid-19," *J. Sci. Technol.*, vol. 11, no. 2, pp. 339–350, 2022, doi: 10.23887/jstundiksha.v11i2.48365.

[33] Desyaf Putra, Dinar Mutiara Kusumo Nugraheni, and Jatmiko Endro Suseno, "Desain Arsitektur Enterprise Berbasis Risiko Teknologi Informasi untuk Sistem Pelatihan: Integrasi COBIT 2019 dan TOGAF ADM," *JST (Jurnal Sains dan Teknol.*, vol. 14, no. 1, pp. 35–46, 2025, doi: 10.23887/jstundiksha.v14i1.93498.

[34] C. Hsu and T. Ohio, "The Delphi Technique :," vol. 12, no. 10, 2007.

This article is licensed under a Creative Commons Attribution- ShareAlike 4.0 International License.

Received: October 18, 2021. Revised: December 20, 2021.21

[35] S. S. Hussein, M. N. ri Mahrin, N. Maarop, and N. A. A. Bakar, "Development and validation of enterprise architecture (EA) readiness assessment model," *Int. J. Adv. Sci. Eng. Inf. Technol.*, vol. 10, no. 1, pp. 157–163, 2020, doi: 10.18517/ijaseit.10.1.9007.

[36] A. T. Jebb, V. Ng, and L. Tay, "A Review of Key Likert Scale Development Advances: 1995–2019," *Front. Psychol.*, vol. 12, no. May, pp. 1–14, 2021, doi: 10.3389/fpsyg.2021.637547.

[37] N. Ilyana *et al.*, "Content Valid ity Assessment Using Aiken ' s V : Knowledge Integration Model for Blockchain in Higher Learning Institutions," vol. 16, no. 6, pp. 601–608, 2025.

[38] L. R. Aiken, "Three Coefficients for Analayzing The Reliability and Validity of Rating," *Sage J.*, vol. 4, no. 1, 1985, [Online]. Available: https://journals.sagepub.com/doi/10.1177/0013164485451012

This article is licensed under a Creative Commons Attribution- ShareAlike 4.0 International License.